

Final Year Project Proposal

Title: Intelligent Web Vulnerability Scanner with Interactive Reporting Dashboard

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Abstract

The increasing reliance on web applications across modern organizations has significantly expanded the attack surface for cyber threats. Many web applications are deployed without comprehensive security assessments, resulting in vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), insecure configurations, and weak authentication mechanisms. This project proposes the development of an Intelligent Web Vulnerability Scanner with an Interactive Reporting Dashboard capable of automatically identifying common web application vulnerabilities through automated crawling, payload injection, and response analysis. The system will classify vulnerabilities according to severity, provide remediation guidance, maintain historical scan records, and present analytical results through a user-friendly dashboard. The proposed platform aims to provide an educational, lightweight, and cost-effective alternative to enterprise vulnerability scanning tools while enhancing practical understanding of web application security assessment.

1. Introduction

Web applications have become integral to digital transformation in sectors including business, healthcare, education, finance, and government. However, rapid development cycles and insufficient secure coding practices frequently introduce exploitable vulnerabilities into deployed applications. Attackers leverage these weaknesses to compromise confidentiality, integrity, and availability of systems.

Automated vulnerability scanning tools play a critical role in identifying security weaknesses before exploitation occurs. Despite the availability of enterprise-grade scanners, their complexity and cost make them inaccessible for many educational institutions, students, and small-scale developers. This project addresses that gap by proposing a lightweight yet technically robust vulnerability scanner designed for academic and practical cybersecurity use.

2. Problem Statement

A significant number of web applications are deployed without adequate vulnerability assessment due to limited access to affordable and user-friendly security testing tools. Existing professional vulnerability scanners such as Burp Suite Professional, Nessus, and Acunetix are often financially inaccessible and operationally complex for beginner users. Consequently, there exists a need for a simplified yet effective vulnerability scanning platform that enables students, developers, and small organizations to identify and understand common web application security flaws.

3. Project Objectives

The major objectives of the proposed system are as follows:

- To design and implement an intelligent automated vulnerability scanner for web applications.
 - To detect common OWASP Top 10 vulnerabilities including:
 - SQL Injection (SQLi)
 - Cross-Site Scripting (XSS)
 - Security Header Misconfigurations
 - Weak Authentication Mechanisms
 - To classify detected vulnerabilities based on severity levels (Low, Medium, High).
 - To provide remediation recommendations for each identified vulnerability.
 - To store and maintain historical scan records for comparative security analysis.
 - To visualize scanning outcomes using an interactive analytical dashboard.
 - To develop a modular and extensible architecture for future security modules.
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4. Scope of the Project

The proposed system focuses specifically on automated security assessment of web applications. It includes URL crawling, input field extraction, payload-based testing, response analysis, vulnerability classification, and dashboard reporting.

Project Boundaries / Limitations

- The scanner will focus on common and known vulnerabilities only.
 - Detection of zero-day vulnerabilities is outside the project scope.
 - Full penetration testing and exploit chaining are excluded.
 - The tool is intended strictly for authorized testing environments.
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5. Proposed Methodology

The implementation methodology of the proposed system consists of the following phases:

Phase 1: Target Submission

The user provides the target web application URL through the system interface.

Phase 2: Automated Crawling

The crawler extracts internal links, forms, parameters, and input fields from the target application.

Phase 3: Payload Injection and Security Testing

Predefined payloads are injected into identified parameters and forms to test for known vulnerabilities.

Phase 4: Response Validation and Analysis

HTTP responses are analyzed to validate whether payloads successfully trigger vulnerability indicators.

Phase 5: Vulnerability Classification

Confirmed findings are categorized according to severity and risk level.

Phase 6: Reporting and Visualization

Results are stored, visualized on the dashboard, and presented in structured report format.

6. Tools and Technologies

Programming Language

- Python

Backend / Scanner Development Libraries

- Requests
- BeautifulSoup
- Selenium (Optional for dynamic web pages)
- Pandas / NumPy

Frameworks

- Flask / Django

Database Systems

- SQLite / MongoDB

Frontend Technologies

- HTML / CSS / JavaScript
- Chart.js / Plotly

Development Environment

- VS Code / PyCharm
 - Git / GitHub
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7. System Architecture

The proposed system follows the architecture below:

User Interface → Crawling Module → Vulnerability Scanner Engine → Response Analysis Module → Database Storage → Dashboard & Reporting Module

8. Expected Outcomes

Upon successful completion, the project is expected to deliver:

- A fully functional automated web vulnerability scanner.
 - Detection and classification of common web application vulnerabilities.
 - Interactive dashboard for visualizing scan metrics and trends.
 - Historical scan tracking and comparative analysis.
 - Actionable remediation suggestions for detected weaknesses.
 - A practical educational cybersecurity assessment platform.
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9. Novelty and Innovation

The proposed project introduces several innovative features beyond basic academic vulnerability scanners:

- Integration of scanning and analytical dashboard within one platform.
 - Automated remediation recommendation engine.
 - Historical vulnerability trend monitoring.
 - Severity-based prioritization of vulnerabilities.
 - Modular framework enabling future expansion and plugin support.
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10. Future Enhancements

Future versions of the system may include:

- Machine Learning-based anomaly detection for advanced vulnerability identification.
 - Authentication-aware scanning for protected web applications.
 - REST API and GraphQL security testing support.
 - Cloud deployment for multi-user accessibility.
 - Exportable PDF and compliance audit reports.
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11. Conclusion

The Intelligent Web Vulnerability Scanner with Interactive Reporting Dashboard provides a practical and educational solution for automated web application security assessment. By integrating vulnerability detection, severity classification, remediation guidance, and dashboard visualization, the system bridges the gap between expensive enterprise-grade tools and academic cybersecurity learning platforms. This project will contribute both educational and practical value by enhancing understanding of secure web application assessment methodologies.

12. References

- OWASP Top 10 Project Documentation
- OWASP Web Security Testing Guide (WSTG)
- Python Official Documentation
- Burp Suite Documentation
- Nessus Documentation
- Acunetix Security Resources